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F–404

B.C.A. (First Semester) Examination, 2018

(New Course)

Paper-No. BCA-104-N

MATHEMATICS-I

Time : 3 Hours] [Maximum Marks : 70

Note :- Attempt any *five* questions. All questions carry equal marks. Symbols used have their usual meanings.

1. (a) Determine the values of a, b, c for which the function :

$$f(x) = \begin{cases} \dfrac{\sin(a+1)x + \sin x}{x} & : x < 0 \\ c & : x = 0 \\ \dfrac{(x+bx^2)^{1/2} - x^{1/2}}{bx^{3/2}} & : x > 0 \end{cases}$$

is continuous at $x = 0$.

(b) Examine the function defined below for continuity at $x = 0$:

$$f(x) = \begin{cases} \dfrac{\sin^2 ax}{x^2} & : x \neq 0 \\ 1 & : x = 0 \end{cases}$$

2. (a) If :

$$I_n = \frac{d^n}{dx^n}(x^n \log x)$$

prove that $I_n = nI_{n-1} + (n-1)!$

(b) If $y = a\cos(\log x) + b\sin(\log x)$, show that :

$$x^2 y_{n+2} + (2n+1)xy_{n+1} + (x^2+1)y_n = 0.$$

3. (a) If $u = e^{xyz}$.

show that :

$$\frac{\partial^3 u}{\partial x \partial y \partial z} = (1 + 3xyz + x^2y^2z^2)e^{xyz}$$

(b) If $(\tan x)^y + y^{\cos x} = a$.

find the value of $\dfrac{dy}{dx}$.

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4. (a) If :

$$x = r \sin\theta \cos\phi$$

$$y = r \sin\theta \sin\phi$$

$$z = r \cos\theta.$$

show that $\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = r^2 \sin\theta$

(b) Evaluate :

$$\int_0^{\pi/2} \log \sin x \, dx$$

5. (a) Obtain the reduction formula for :

$$\int \sec^n x \, dx$$

(b) Find the area bounded by the parabola $y^2 = 4ax$ and its latus rectum

6. (a) Find the length of the astroid :

$$x^{2/3} + y^{2/3} = a^{2/3}$$

(b) Show that the volume of a sphere of radius a is $\frac{4}{3}\pi a^3$.

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7. (a) Evaluate grad r^n.

(b) If $r = a\cos t\,\hat{i} + a\sin t\,\hat{j} + at\tan\alpha\,\hat{k}$, find :

$$\left|\frac{dr}{dt} \times \frac{d^2r}{dt^2}\right|$$

8. (a) Solve the differential equation :

$$(D^4 + m^4)y = 0$$

(b) Solve :

$$(D^2 + a^2)y = \sec ax$$

9. (a) Find the equation of the hyperbola whose focus is (1, 1), directrix $2x + 2y = 1$ and eccentricity is $\sqrt{2}$.

(b) Find the vertex, focus and directrix of the parabola :

$$x^2 + 8x + 12y + 4 = 0$$

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